CSE 4711: Artificial Intelligence - Quiz #1 **29 August 2023**
Duration: 20 Minutes **Full Marks: 15**
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Imagine a variant of the game Super Mario, where the task of Mario is to navigate a grid of size $X \times Y$. The grid contains C coins fixed in specific positions. Starting from the bottom-left corner, Mario's objective is to collect all coins and drop them at the coin flag at the top-right corner. At each timestep, Mario can choose an action: turning north, east, south, or west, moving forward, or staying still. Each action incurs a cost of 1. If moving forward means falling off the grid, the action is not allowed. Mario can carry up to $M\,(1 \leq M \leq C)$ coins simultaneously. When entering a square containing a coin, he takes it if he is not already carrying M coins; otherwise, the coin will remain in the cell. Upon reaching the coin flag, Mario automatically drops all the coins he's carrying. The game concludes when all coins are collected and deposited at the coin flag. A sample 3×4 grid is shown in Figure 1.

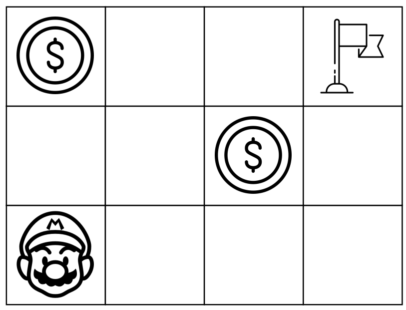

Figure 1: A sample grid showing a variant of the game Super Mario

Consider that you came up with 3 heuristics, h_1, h_2, and h_3 applicable to the problem. Among these, h_1 is admissible, h_2 is consistent, and h_3 lacks both admissibility and consistency. However, when integrated into an A* graph search, h_3 is guaranteed to lead to an optimal solution for the given problem.

1. Formulate the scenario as a search problem by identifying the features stored for each state, actions, start state, and goal test. 4 (CO2) (PO2)

> **Solution:**
>
> **Features:** The coordinates of Mario, the direction Mario is facing, Number of coins currently carried by Mario, and Coin Boolean.
> **Actions:** Turn North, Turn East, Turn South, Turn West, Move Forward, Stay Still
> **Start State:** The initial game configuration
> **Goal Test:** All values of Coin Boolean are False and Mario is in the Top-Right corner.
>
> **Rubric:**
>
> - 1 point for each portion

2. With a brief justification, recommend an uninformed search algorithm that will find the optimal solution. 1 + 1 (CO3) (PO3)

> **Solution:**
>
> Breadth-First Search/Iterative Deepening/Uniform Cost Search

Since each action costs 1, Breadth-First Search/Iterative Deepening/Uniform Cost Search will provide the optimal solution.

Rubric:

- 1 point for algorithm choice.
- 1 points for justification.

3. In addition to the given scenario, let there are G Goombas present on the grid. Each Goomba has the following set of actions: at each timestep, they move forward given that it does not result in falling off the grid; otherwise, they turn right. Goombas can move over each other. These actions are deterministic and known to Mario, occurring at the same time as Mario's actions. That means when Mario makes a move, all the Goombas make a move. If Mario and a Goomba happen to be in the same cell, Mario loses all his collected coins that have not been deposited yet, which are then restored to their original locations. At the same time, Mario is moved to his starting position. 3×3 (CO3) (PO3)

   With proper justification, comment on the following statements considering the new scenario:

   a) h_1 will remain admissible.

   Solution:
   Yes.
   The revised problem can be perceived as employing the identical state space as the original. However, it incorporates extra arcs that trace back from states nearer to the goal to those that are more distant (when Mario's coins are forfeited and reset). In this context, h_1 remains admissible, as the altered problem formulation does not introduce any connections that shorten the path to the goal.

   Rubric:

   - 1 point for decision.
   - 2 points for justification.

   b) h_2 will remain consistent.

   Solution:
   Yes.
   The revised problem can be perceived as employing the identical state space as the original. However, it incorporates extra arcs that trace back from states nearer to the goal to those that are more distant (when Mario's coins are forfeited and reset). In this context, h_2 remains consistent, as the altered problem formulation does not introduce any connections that shorten the path to the goal.

   Rubric:

   - 1 point for decision.
   - 2 points for justification.

   c) When integrated into an A* graph search, h_3 is guaranteed to lead to an optimal solution.

Solution:

No.

Let h_3 be a heuristic that provides lower estimates for the states along the optimal path to the goal in the initial problem. However, for all other states, it gives higher estimates. Hence, it is not admissible for the original problem. Introducing Goombas might cause the optimal path to pass through various states with a higher estimated distance from the goal in the revised problem. This can potentially hinder the exploration of those states during the search process.

Rubric:

- 1 point for decision.
- 2 points for justification.